Decomposition Methods in Maximum Satisfiability: Connecting Core-Guided and Implicit Hitting Set Algorithms with Benders and Dantzig–Wolfe Decomposition

Technical Report

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Abstract

Maximum Satisfiability (MaxSAT) and Mixed-Integer Programming (MIP) are two major frameworks for combinatorial optimization that have largely developed in separate communities. In this report we draw explicit structural parallels between two families of MaxSAT algorithms and two classical decomposition methods from operations research. We show that core-guided MaxSAT algorithms operate as instances of Logic-Based Benders Decomposition (LBBD), where unsatisfiable cores play the role of Benders cuts derived via inference duality. Dually, we argue that Implicit Hitting Set (IHS) MaxSAT solvers implement a form of Dantzig–Wolfe decomposition when viewed through LP duality, with the SAT oracle acting as a pricing subproblem that generates new constraints (columns in the dual). These connections unify the algorithmic landscape across the two communities and suggest avenues for cross-fertilization.

1 Introduction

Maximum Satisfiability (MaxSAT) is the optimization version of the Boolean Satisfiability problem (SAT): given a set of hard clauses $\mathcal{H}$ and a set of weighted soft clauses $(\mathcal{S}, w)$, find a truth assignment that satisfies all hard clauses while maximizing the total weight of satisfied soft clauses (equivalently, minimizing the cost of falsified soft clauses).

Modern exact MaxSAT solvers fall into two broad families:

1. **Core-guided algorithms** (e.g., WPM1 [Ansotegui et al., 2009], PMRES [Narodytska and Bacchus, 2014], OLL [Morgado et al., 2014]), which iteratively call a SAT solver, extract unsatisfiable cores, and transform the formula to raise a lower bound.

2. **Implicit Hitting Set (IHS) algorithms** (e.g., MaxHS [Davies and Bacchus, 2011, 2013]), which alternate between a SAT oracle that extracts cores and an IP solver that computes a minimum-cost hitting set of the cores found so far.

In operations research, Benders decomposition [Benders, 1962] and Dantzig–Wolfe decomposition [Dantzig and Wolfe, 1960] are two classical methods for structured optimization problems. Both decompose a hard monolithic problem into a coordinating master problem and simpler subproblems, iterating by passing cuts or columns between them. They are linked through LP duality: Benders adds *rows* (cuts) to a master problem defined over complicating variables, while Dantzig–Wolfe adds *columns* to a master problem defined over complicating constraints.

The purpose of this report is to make explicit the structural correspondences:

- Core-guided MaxSAT $\longleftrightarrow$ Logic-Based Benders Decomposition,
- IHS MaxSAT $\longleftrightarrow$ Dantzig–Wolfe Decomposition (in the dual).

2 Preliminaries

2.1 MaxSAT

Definition 1 (Weighted Partial MaxSAT)**.** *An instance is a triple $\langle \mathcal{H}, \mathcal{S}, w \rangle$ where $\mathcal{H}$ is a set of hard clauses, $\mathcal{S} = \{s_1, \ldots, s_m\}$ is a set of soft clauses, and $w : \mathcal{S} \to \mathbb{Q}_{>0}$ assigns weights. A solution is an assignment σ satisfying all clauses in $\mathcal{H}$; its cost is $\sum_{s_i : \sigma \not\models s_i} w(s_i)$. The goal is to find a minimum-cost solution.*

Definition 2 (Core)**.** *A* core *of $\langle \mathcal{H}, \mathcal{S}, w \rangle$ is a subset $\kappa \subseteq \mathcal{S}$ such that $\mathcal{H} \cup \kappa$ is unsatisfiable. A core is* minimal *(a MUS) if no proper subset is also a core.*

Definition 3 (Correction Set)**.** *A* correction set *is a subset $\mathcal{C} \subseteq \mathcal{S}$ whose removal makes the formula satisfiable, i.e., $\mathcal{H} \cup (\mathcal{S} \setminus \mathcal{C})$ is satisfiable. A correction set is* minimal *(an MCS) if no proper subset has this property.*

There is a well-known duality: every minimal core hits every minimal correction set, and vice versa [Liffiton and Sakallah, 2008].

2.2 The LP Relaxation of MaxSAT

Introduce a binary variable $y_i \in \{0, 1\}$ for each soft clause s_i, where $y_i = 1$ means s_i is falsified. The MaxSAT problem can be written as the integer program:

$$\min \quad \sum_{i=1}^{m} w(s_i)\, y_i \tag{1}$$

$$\text{s.t.} \quad \sum_{s_i \in \kappa} y_i \geq 1 \quad \forall \text{ cores } \kappa, \tag{2}$$

$$y_i \in \{0, 1\}. \tag{3}$$

Constraint (2) states that every core must be "hit" by at least one falsified soft clause. This is the *hitting set* formulation, and there are exponentially many core constraints. The LP relaxation replaces (3) with $0 \leq y_i \leq 1$.

2.3 Benders Decomposition

Classical Benders decomposition [Benders, 1962] partitions variables into complicating variables x and easy variables y. The master problem optimizes over x; for each fixed $\bar{x}$, a subproblem over y either certifies feasibility or returns a *Benders cut* derived from the subproblem's dual. Cuts are accumulated in the master.

Logic-Based Benders Decomposition (LBBD) [Hooker and Ottosson, 2003] generalizes this by allowing the subproblem to be any optimization or feasibility problem. Cuts are derived via an *inference dual*: any proof of infeasibility or suboptimality of the subproblem yields a valid cut for the master. The cut need not come from LP duality; it can come from any sound deduction system.

2.4 Dantzig–Wolfe Decomposition

Dantzig–Wolfe decomposition [Dantzig and Wolfe, 1960] addresses problems with complicating *constraints*. The feasible region of a subproblem is represented via its extreme points. A restricted master problem uses a subset of these extreme points (columns); a *pricing subproblem* identifies whether any extreme point with negative reduced cost exists. If so, the corresponding column is added to the master. The procedure terminates when no such column exists.

Through LP duality, Benders and Dantzig–Wolfe are intimately related: adding a row (cut) to the primal master is equivalent to adding a column to the dual master, and vice versa.

3 Core-Guided MaxSAT as Benders Decomposition

3.1 The Decomposition Structure

A core-guided MaxSAT algorithm operates in the following loop:

1. **Subproblem (SAT oracle):** Given the current (transformed) formula, call a SAT solver. If satisfiable, the model is optimal; stop. If unsatisfiable, obtain a core κ.
2. **Master update:** Use the core κ to transform the formula (add relaxation variables, cardinality constraints) so that the lower bound increases.
3. Repeat.

We now map this onto the LBBD framework of Hooker and Ottosson [2003].

LBBD component	**Core-guided MaxSAT analog**
Complicating variables x	Soft clause falsification decisions y_i
Master problem	Accumulated lower bound + constraints
Subproblem SP($\bar{x}$)	SAT call on $\mathcal{H} \cup \mathcal{S}'$
Infeasibility certificate	Unsatisfiable core κ
Benders cut	Hitting set constraint: $\sum_{s_i \in \kappa} y_i \geq 1$
Inference dual	SAT proof of unsatisfiability (e.g., resolution)

3.2 The Inference Dual Perspective

In classical Benders, the cut $\pi^T b \leq z$ is derived from the LP dual of the subproblem. In LBBD, the cut comes from any valid proof. In core-guided MaxSAT, the SAT solver provides a refutation proof (via conflict-driven clause learning) that establishes the unsatisfiability of $\mathcal{H} \cup \kappa$. This proof serves as the solution to the inference dual: it certifies that *at least one* soft clause in κ must be falsified, producing the cut

$$\sum_{s_i \in \kappa} y_i \geq 1.$$

This is a *combinatorial Benders cut* [Hooker and Ottosson, 2003, Codato and Fischetti, 2006]: a linear inequality derived from a logical deduction rather than from LP duality.

3.3 Formula Transformation as Implicit Cut Accumulation

Core-guided solvers such as PMRES and OLL do not maintain an explicit master IP. Instead, they transform the formula by adding relaxation variables and cardinality constraints. Recent work by Katsirelos [2023] has shown that these transformations implicitly encode an LP. Specifically:

Proposition 1 (Katsirelos, 2023)**.** *The transformations performed by PMRES and OLL implicitly discover cores of the original formula. In unweighted instances, the relaxed formula at each iteration encodes a minimum hitting set of these cores. Moreover, the process can be captured by a compact integer linear program whose LP relaxation yields the same bound.*

Thus, what appears to be a purely syntactic formula transformation is, in fact, an implicit construction of the Benders master problem, with each core extraction step corresponding to one iteration of the LBBD loop.

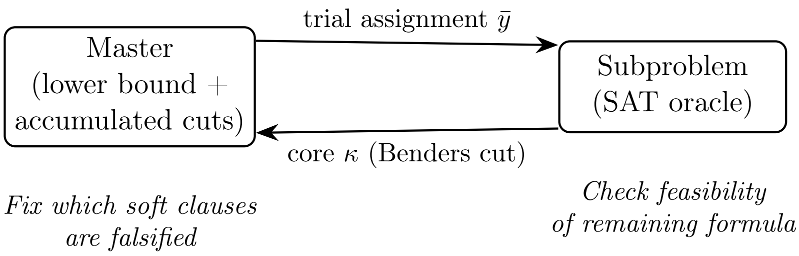

Figure 1: Core-guided MaxSAT as Logic-Based Benders Decomposition. The SAT oracle acts as the subproblem; each unsatisfiable core yields a combinatorial Benders cut.

4 IHS MaxSAT as Dantzig–Wolfe Decomposition

4.1 The Implicit Hitting Set Framework

The IHS algorithm [Davies and Bacchus, 2011, Moreno-Centeno and Karp, 2013] operates as follows:

1. **Master problem (IP solver):** Compute a minimum-cost hitting set H^* of all cores found so far. That is, solve

$$\min \sum_{i=1}^{m} w(s_i)\, y_i \quad \text{s.t.} \quad \sum_{s_i \in \kappa_j} y_i \geq 1 \;\; \forall j = 1, \ldots, k, \quad y_i \in \{0, 1\}.$$

2. **Oracle (SAT solver):** Check whether removing the clauses in H^* from $\mathcal{S}$ makes the formula satisfiable. If yes, H^* is optimal; stop. If not, the SAT solver returns a new core κ_{k+1} that is not hit by H^*.

3. Add κ_{k+1} as a new constraint and repeat.

4.2 The Dantzig–Wolfe Connection via LP Duality

To see the Dantzig–Wolfe structure, consider the LP dual of the hitting set formulation (1)–(2). Let $\mathcal{K} = \{\kappa_1, \ldots, \kappa_N\}$ be the set of all cores, and let $\lambda_j \geq 0$ be the dual variable for core κ_j. The dual LP is:

$$\max \quad \sum_{j=1}^{N} \lambda_j \tag{4}$$

$$\text{s.t.} \quad \sum_{j\,:\,s_i \in \kappa_j} \lambda_j \leq w(s_i) \quad \forall i = 1, \ldots, m, \tag{5}$$

$$\lambda_j \geq 0. \tag{6}$$

This is a *packing* problem over cores: find a maximum-weight fractional packing of cores subject to the capacity of each soft clause.

Since the number of cores N is exponential, this dual cannot be solved directly. However, it has the structure required for *column generation* (the algorithmic engine of Dantzig–Wolfe decomposition):

- The **restricted master** is the dual LP (4)–(6) restricted to the cores $\kappa_1, \ldots, \kappa_k$ discovered so far. Each core κ_j corresponds to a *column* (variable λ_j) in this dual.

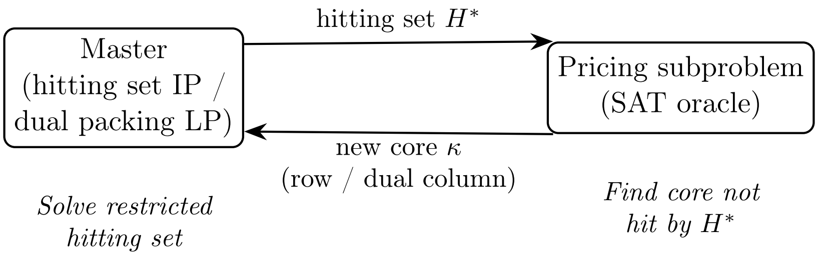

Figure 2: IHS MaxSAT as Dantzig–Wolfe Decomposition (dual view). The SAT oracle acts as the pricing subproblem; each new core is a column in the dual packing LP.

- The **pricing subproblem** asks: does there exist a core κ not in the current restricted master that has positive reduced cost? In the dual (4)–(6), this amounts to finding a core κ such that $\mathcal{H} \cup \kappa$ is unsatisfiable and κ is not hit by the current primal solution. The SAT oracle answers exactly this question.

Dantzig–Wolfe component	**IHS MaxSAT analog**
Restricted master (dual)	Packing LP over known cores
Column	Core κ_j (dual variable λ_j)
Pricing subproblem	SAT oracle: find unhit core
Negative reduced cost	Core not hit by current hitting set
Restricted master (primal)	Hitting set IP over known cores
New constraint (row)	New core κ_{k+1}

Proposition 2. *The IHS algorithm is a primal-side implementation of column generation on the dual of the MaxSAT hitting set LP. Each new core discovered by the SAT oracle acts simultaneously as:*

1. *a new* row *(constraint) in the primal hitting set master, and*
2. *a new* column *(variable) in the dual packing master.*

By LP duality, adding a row to the primal is equivalent to adding a column to the dual. Hence the IHS algorithm is a Dantzig–Wolfe procedure on the dual formulation.

4.3 LP Bounds and Convergence

In Dantzig–Wolfe decomposition, the restricted master LP provides a bound that improves monotonically as columns are added, converging to the LP optimum. Similarly, in IHS, adding more cores tightens the LP relaxation of the hitting set master. Davies and Bacchus [2013] showed that solving the hitting set master as an IP (rather than an LP) accelerates convergence in practice by providing both primal feasible solutions and dual bounds simultaneously.

The work of Katsirelos [2023] further showed that core-guided algorithms such as OLL implicitly compute feasible dual solutions of an LP that encodes the same hitting set structure. This means that both families of algorithms—core-guided and IHS—are optimizing over the *same* LP relaxation, but from opposite sides of duality.

5 The Unified Picture

The preceding analysis reveals a clean duality diagram connecting the four methods:

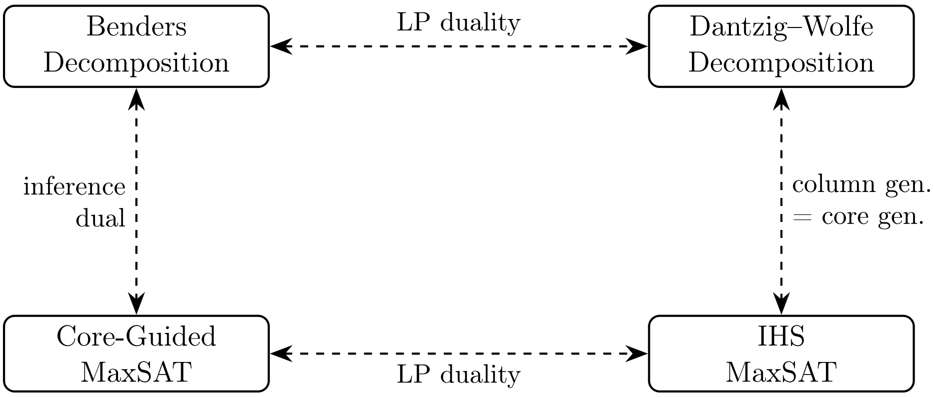

Figure 3: The duality square connecting decomposition methods and MaxSAT algorithms.

- **Horizontal axis (LP duality):** Benders and Dantzig–Wolfe are dual to each other; core-guided and IHS are dual to each other in the same sense. Adding a cut in one is adding a column in the other.
- **Vertical axis (generalization):** Core-guided MaxSAT is an instance of LBBD where the inference dual replaces the LP dual. IHS MaxSAT is an instance of Dantzig–Wolfe where the pricing oracle is a SAT solver rather than an LP.

This diagram also explains empirically observed complementarities. IHS solvers (like Dantzig–Wolfe) tend to produce good primal solutions quickly because the master IP yields feasible hitting sets at each iteration. Core-guided solvers (like Benders) tend to produce strong dual bounds because each core directly tightens the relaxation. This mirrors the well-known tradeoff in OR between cut-based and column-based approaches.

6 Discussion

Implications for algorithm design. The decomposition viewpoint suggests several ideas for improving MaxSAT solvers:

1. *Warm-starting*: In OR, Benders and Dantzig–Wolfe solvers routinely warm-start the master with cuts or columns from heuristic solutions. The same idea applies to MaxSAT: seeding the IHS master with cores from a quick core-guided run, or vice versa.
2. *Cut strengthening*: The Benders literature contains extensive work on strengthening cuts (Pareto-optimal cuts, Magnanti–Wong cuts). Analogous techniques could be applied to core minimization and core-guided clause transformations.
3. *Stabilization*: Column generation is notoriously unstable (the "tailing-off" effect). Stabilization techniques (dual penalties, bundle methods) could improve IHS convergence.
4. *Hybrid methods*: The recent SAT 2025 work of Katsirelos [2025] shows that OLL can be restated as an algorithm that explicitly computes dual solutions of an LP, opening the door to hybrid algorithms that interleave core-guided (Benders-like) and IHS (Dantzig–Wolfe-like) steps within the same solve.

Limitations. The analogy is structural, not exact. Core-guided solvers work with integer formulas and perform syntactic transformations, not LP pivots. The IHS master is typically solved as an IP, not an LP, which goes beyond standard column generation. Moreover, the subproblem in both cases is a SAT call—an NP-complete problem—rather than the polynomial-time LP solve assumed in classical decomposition theory. Nonetheless, the structural parallels are useful for transferring insights between communities.

7 Conclusion

We have presented explicit structural correspondences between two families of MaxSAT algorithms and two classical decomposition methods from operations research. Core-guided MaxSAT algorithms implement a form of Logic-Based Benders Decomposition, where unsatisfiable cores serve as combinatorial Benders cuts derived through inference duality. Implicit Hitting Set algorithms implement a form of Dantzig–Wolfe decomposition (viewed through LP duality), where the SAT oracle acts as a pricing subproblem generating new columns in the dual packing formulation. These connections are mediated by LP duality: the hitting set (primal) and core packing (dual) formulations of MaxSAT stand in the same relationship as Benders and Dantzig–Wolfe masters.

Making these connections explicit opens a two-way transfer of techniques between the OR and SAT communities, from cut strengthening and stabilization to hybrid primal-dual algorithms that leverage the strengths of both decomposition paradigms.

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